



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



Department of Computer Science and Engineering(Data Science)

SMASHIFIX : AN APPLICATION FOR BADMINTON POSTURE CORRECTION USING DEEP LEARNING

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Introduction

- Computer vision and AI have revolutionized sports analysis by enabling precise, real-time tracking of player performance and biomechanics.
- In badminton, where footwork, stroke technique, and positioning are highly complex, these systems provide accuracy beyond human observation.
- Automated computer-vision-based systems can analyze posture and movement frame by frame, offering insights critical for player development.
- Deep learning integration allows precise detection of intricate body mechanics for training and competition analysis.
- Pose estimation algorithms map body positions using part affinity fields to study player movements in detail.
- Trajectory-based analyses track movement patterns to profile tactics and gameplay strategies.
- These methods surpass traditional observation by providing data-driven feedback for both technical and strategic improvement.
- Ongoing advancements in validation and system development continue to enhance badminton training and competitive performance.

Literature Survey

No	Name of Paper	Conference / Journal (Year)	Methodologies	Parameters	Gaps
1.	Tennis player action dataset for human pose estimation	"Science Direct" (2024)	• Curation of Dataset using labelled data • COCO-Annotation used to highlight skeletal keypoints. • Vid frame extraction using smartphone's cam.	• 17 keypoints annotated using COCO-Annotator • Motion analysis, Injury prevention an perf. Improvement.	• Lighting variations. • Camera and background noise. • Non -inclusion of more diverse players. • Lack of 3D pose estimation. • Lack of multiview vid data.
2.	Research on Badminton Teaching Technology Based on Human Pose Estimation Algorithm	"Machine Learning and Scientific Programing in Multi-Sensor Data Processing" (2022)	• Data Collection and Annotation. • Real-Time Feedback: • Convolutional Neural Networks (CNNs). • Multi-Person Pose Estimation. • Part Confidence Map.	• A comparison between two groups: one using the proposed method and the other using traditional teaching methods . • Contact point, stroke trajectory, stroke score. • Statistical analysis methods.	• Availability of dedicated datasets for badminton stance analysis • Constructed dataset only includes two skill movements.
3.	Application of Human Posture Recognition Based on the Convolutional Neural Network in Physical Training Guidance	"Computational Intelligence and Neuroscience" (2022)	• Convolutional Network Overview • Feature Map Size Maintenance • Stacked Hourglass Architecture • Large Receptive Field Module	• Addressing Errors in Complex Sports Video Environments • Superimposed Hourglass Grid. • The Need for Multimodal Approaches in Athlete Posture	• Only estimates the pose of a single athlete in the sports game video • It has shortcomings in practical application and needs further research.

Literature Survey

No.	Name of Paper	Conference / Journal (Year)	Methodologies	Parameters	Gaps
4.	Machine Learning in Mobile Applications	"International Journal of Computer Science and Mobile Computing " (2022)	• Server-Side ML: Heavy computation is performed in the cloud. • Client-Side ML: Models run on the device for real-time inference, reducing latency. • ML frameworks such as TensorFlow, ML Kit, Pytorch used.	• Computation efficeincy (CPU/GPU Usage) • Srorage requirements • Network Latency, server side and client side repsonse time. • Scalability ,handling multiple requests and updating models	• Latency Issues in Server-Side ML, high infrastructure cost. • Client-Side ML – Model updates require app updates, making maintenance difficult. • Running ML models on mobile can drain battery quickly.
5.	Deep Learning Based Technical Classification of Badminton Pose with Convolutional Neural Networks	"ILKOM Jurnal Ilmiah" (2024)	• Two-Stage Pipeline in the Blaze-Pose Architecture. • Keypoint Regression Method.	• Confusion Matrix for Checking Model Performance. • Using f1-score, precision and recall.	• Limited dataset size and diversity for covering all badminton techniques and scenarios. • Model performance affected by outliers or unusual poses; requires further data preprocessing.
6.	Vision Based Automated Badminton Action Recognition Using the New Local Convolutional Neural Network Extractor	"Enhancing Health and Sports Performance by Design" (2020)	• Halved to reduce empty space and improve accuracy. • AlexNet used for feature extraction via transfer learning ("local CNN").	• Validation: Two confusion matrices used post Local CNN, AlexNet, and SVC. • Accuracy metric assesses improvement with Local CNN.	• Limited Actions; Only hit vs. non-hit. • Small Dataset; 2,990 images from 5 videos; no diversity.

Literature Survey

No.	Name of Paper	Conference / Journal (Year)	Methodologies	Parameters	Gaps
7.	Real-time Posture Correction of Squat Exercise: A Deep Learning Approach for Performance Analysis and Error Correction	"IEEE Access" (2025)	• Stereo camera captures squat videos. • MediaPipe extracts and tracks keypoints. • Bi-CGRU classifies squats. • Deep learning regressor estimates correct joint angles.	• Squat classification types • Joint angles measured • Dataset • Accuracy Metrics - Bi-CGRU model - Pose correction regression model	• Single-camera limitation • Pose estimation noise • Limited dataset diversity • No real-time mobile implementation • Lack of personalized feedback
8.	Enhancing Cricket Performance Analysis with Human Pose Estimation and Machine Learning	"Sensors" (2023)	• Pre-processing removes noise from videos. • MediaPipe extracts 17 keypoints for skeletal pose representation. • Models tested: RF, SVM, KNN, Decision Tree, LR, LSTM.	• Precision, Recall, F1 score • Acc : 97% , K fold : 95% with SD 0.07 • Trained on various 3 different splits to entire generalization and get better understanding of performance	• Lacked defensive and modern reverse shots. • Unbalanced dataset, with more On Drive than Flick shots. • PCA caused minor data loss.
9.	Yoga Pose Detection and Correction using Posnet and KNN	"IRJET" (2022).	• Images optimized to uniform size. • KNN classifier detected poses via limb keypoints. • Webcam provided real-time feed.	• Training Accuracy : 93.56% • Testing Accuracy: 98.51% • Testing using real time feed	• Very less poses used • OpenPose can be inaccurate during overlaps

Research Gaps

1. Most existing research focuses on pose estimation or movement classification, with limited emphasis on real-time corrective feedback.
2. Current correction-based systems (e.g., yoga applications) have limited scope in terms of the number of poses and are unsuitable for complex, fast-paced sports like badminton.
3. Limited mobile accessibility in existing sports training systems restricts user convenience and flexibility.
4. Most research models are limited to a small number of shot types and movement patterns in badminton analysis.

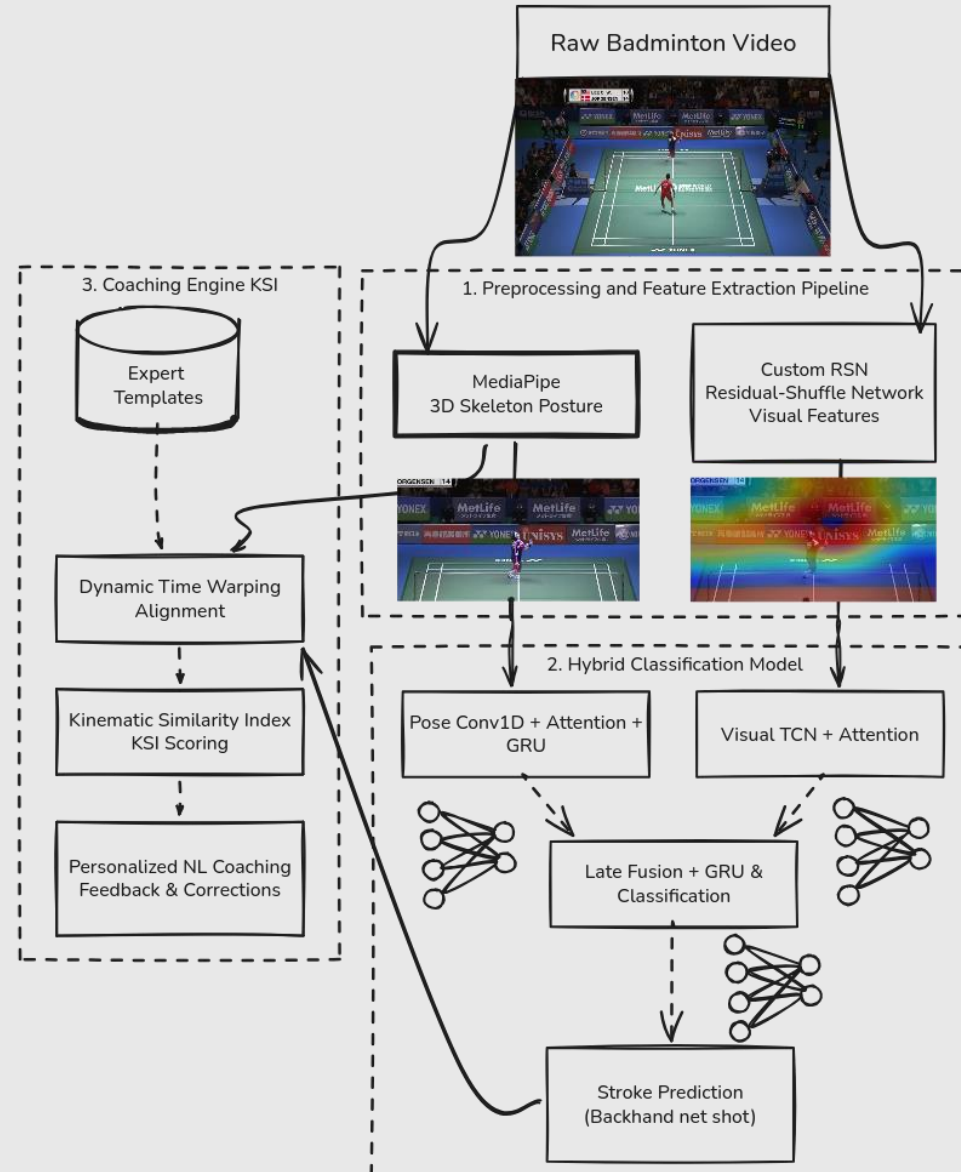


Problem Statement

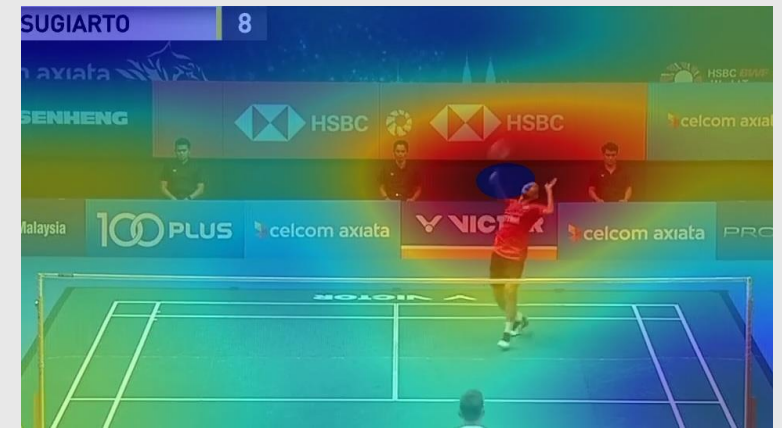
Traditional badminton training relies heavily on manual coaching, which often misses subtle posture errors and limits real-time feedback, ultimately leading to inefficient skill development and a higher risk of injury. To overcome these issues, an automated computer vision-based system utilizes Deep Learning and pose estimation to provide precise, objective, and instant feedback. By detecting incorrect movements during practice and offering real-time visual or text cues, the system corrects techniques on the spot while tracking performance over time to build personalized training plans. Ultimately, bridging the gap between subjective coaching and data-driven precision ensures safer, more effective, and highly scalable player development.



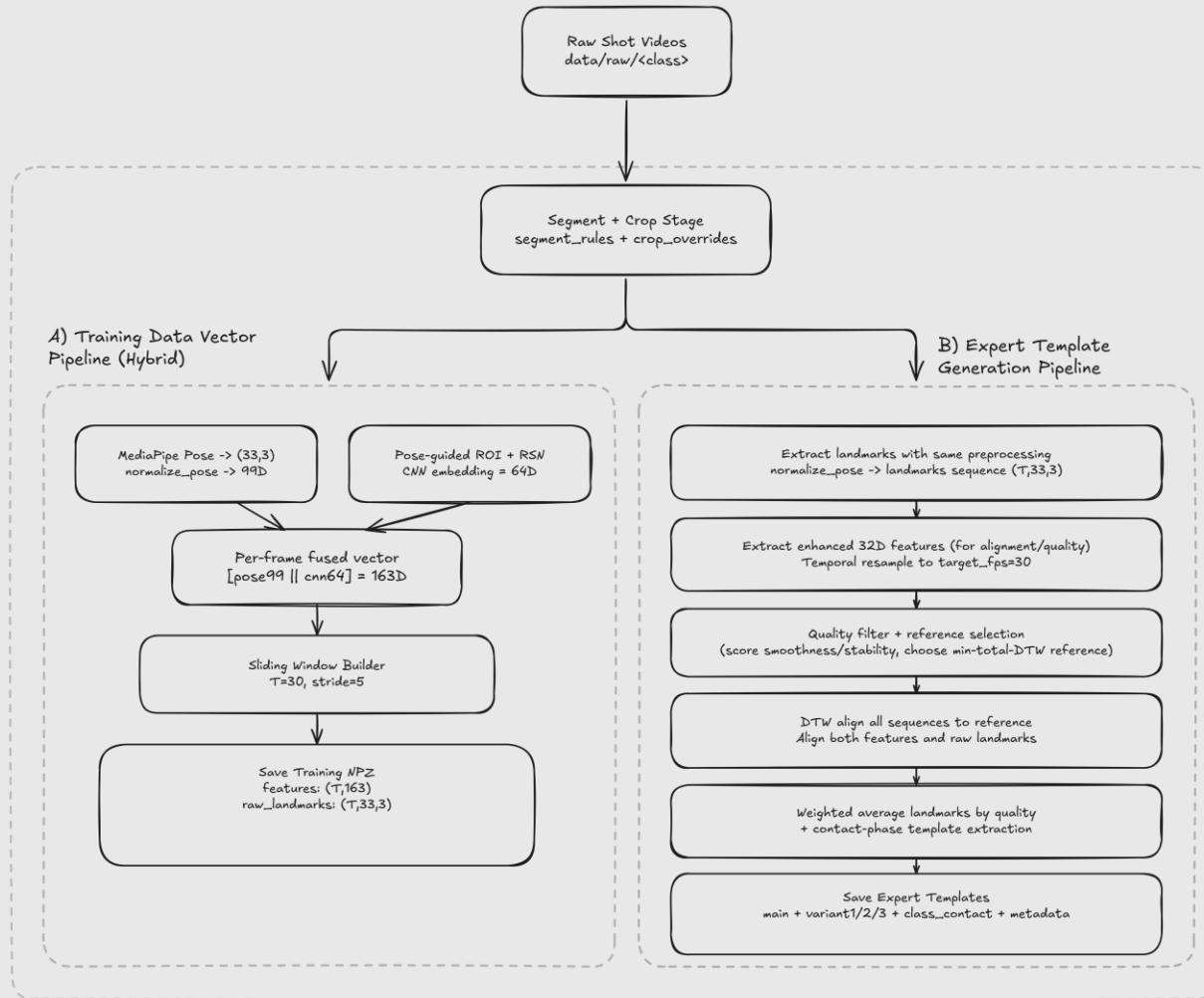
System Architecture & Workflow



Keypoints Extraction and CNN Feature Detection



Feature Extraction and Expert Template Creation

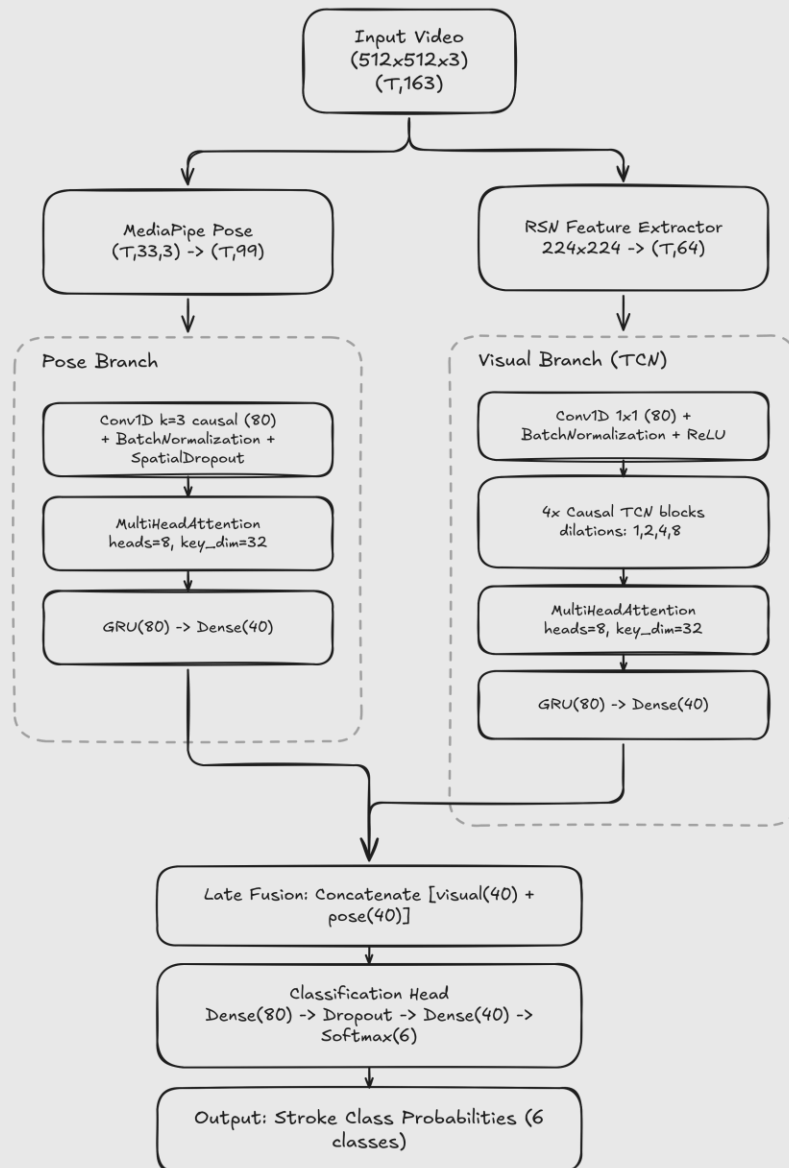


Data Preparation Pipeline

Expert Template Creation:

- I. Standardize inputs: apply same segmentation, crop rules, pose format, and common FPS resampling so all expert videos are directly comparable.
- II. Two motion views: keep (1) normalized landmarks for final templates and (2) compact enhanced features for alignment and quality checks.
- III. Quality filtering: discard low-quality clips using smoothness, stability, and motion-presence criteria to avoid noisy templates.
- IV. Reference selection: choose the most central expert sequence (minimum total DTW distance to others) as alignment anchor.
- V. DTW alignment: temporally align all expert motions to the anchor, preserving phase correspondence despite speed differences.
- VI. Weighted fusion: build the final class template by quality-weighted averaging of aligned landmarks (better clips contribute more).
- VII. Save outputs: store main template, style variants (variant1/2/3), and contact-phase template for sharper fault-focused KSI feedback.

Model Architecture



Hybrid Dual-Stream Architecture

- **Hybrid Dual-Stream Architecture** The model simultaneously processes visual and biomechanical data to maximize shot recognition accuracy:
- **Visual Stream:** Self-trained **Residual-Shuffle Network (RSN)** extracts 64D visual features from the configuration-cropped ROI via channel shuffle operations and residual connections.
- **Attention-TCN** (not just generic TCN) with 4 dilated causal Conv1D layers (dilations 1, 2, 4, 8)
- **Multi-Head Self-Attention** (8 heads, key dim 32) with residual connections
- Optimized hyperparameters: F=80 filters, G=80 GRU units, dropout=0.22, label smoothing=0.1
- **Pose Stream:** A dedicated GRU models 99D skeletal landmarks (33 joints) to track precise body mechanics, independent of background noise.
- **Fusion & Prediction:** Outputs from both streams merge into a final GRU, synthesizing context and mechanics to predict one of 6 shot classes.
- **Sliding Window:** The system captures the full stroke progression by analyzing 40-frame sequences with a 5-frame stride.

Pose Correction : Kinetic Similarity Index

- Step 1: Normalize pose to remove camera/body-scale bias

$$c = \frac{P_{LH} + P_{RH}}{2}, \quad \tilde{P} = P - c, \quad \bar{P} = \frac{P'}{l}$$

- Step 2: Build fused frame-level representation

$$x_t = [\text{vec}(\bar{P}_t) \parallel c_t] \in \mathbb{R}^{163}$$

- Step 3: Contact-aware shot decision (most informative instant)

$$A = \|\Delta v\|_2, \quad \text{Class} = \arg \max_c p(c \mid X_{\text{peak_acc}})$$

- Step 4: Align expert and user sequences with DTW

$$D_{i,j} = \|F_i^E - F_j^U\|_2 + \min\{D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1}\}$$

- Step 5: Give higher importance to key stroke phases

$$\alpha_k = w_{\text{phase}} \left(0.5 + 0.5 \exp \left[-0.5 \left(\frac{k - \text{center}}{\text{width}} \right)^2 \right] \right)$$

One-line takeaway: KSI first normalizes and aligns motion, then evaluates similarity with phase-aware weighting.

Pose Correction : Kinetic Similarity Index

- Pose similarity (shape/biomechanics match)

$$S_{\text{pose}} = \frac{\sum_k \alpha_k \cdot \frac{F_k^E \cdot F_k^U}{\|F_k^E\| \|F_k^U\|}}{\sum_k \alpha_k}$$

- Velocity similarity (timing + speed profile match)

$$S_{\text{vel}} = \frac{\sum_k \alpha_k \cdot \cos_{\text{vel},k} \cdot \exp(-0.1 \|V_k^E - V_k^U\|^2)}{\sum_k \alpha_k}$$

- Final weighted score

$$\text{KSI}_{\text{total}} = w_p S_{\text{pose}} + w_v S_{\text{vel}} + w_a S_{\text{acc}}$$

I. Interpretation:

1. Higher KSI -> user stroke is closer to expert mechanics
2. Lower KSI -> larger deviation in posture, timing, or acceleration

II. Practical use in coaching:

1. Rank performance quality objectively
2. Identify weak phases/joints for correction
3. Generate targeted drill recommendations
4. Track improvement across sessions

Pose Correction : Kinetic Similarity Index

- **Step 1: Compute per-feature error over time (after DTW alignment)**

$$e_{k,t} = |F_{k,t}^E - F_{k,t}^U|$$

where k is biomechanical feature/joint metric and t is aligned frame index.

- **Step 2: Find exact fault frame using phase-weighted error**

$$t_k^* = \arg \max_t (\alpha_t \cdot e_{k,t})$$

- t_k^* gives the most critical frame for each fault (e.g., elbow drop, spine lean).
- DTW path mapping links aligned index back to original user frame/timestamp.

- **Step 3: Rank faults by severity and coaching priority**

- High mean/max error + critical phase (acceleration/contact) -> top priority issue.
- Timing faults use peak-velocity offset (early/late swing) for temporal feedback.

- **Step 4: Convert faults to natural-language coaching**

- Structured mapping: **Issue -> probable cause -> correction cue -> drill.**
- Example style:
 - “Your peak speed occurs late at contact.”
 - “Fix: start forward swing earlier.”
 - “Drill: shadow swings with contact-timing count.”

- **Step 5: Personalized output to user**

- Skill-adaptive wording (beginner/intermediate/advanced/expert)
- Top 3 critical faults + exact frame references + corrective drills

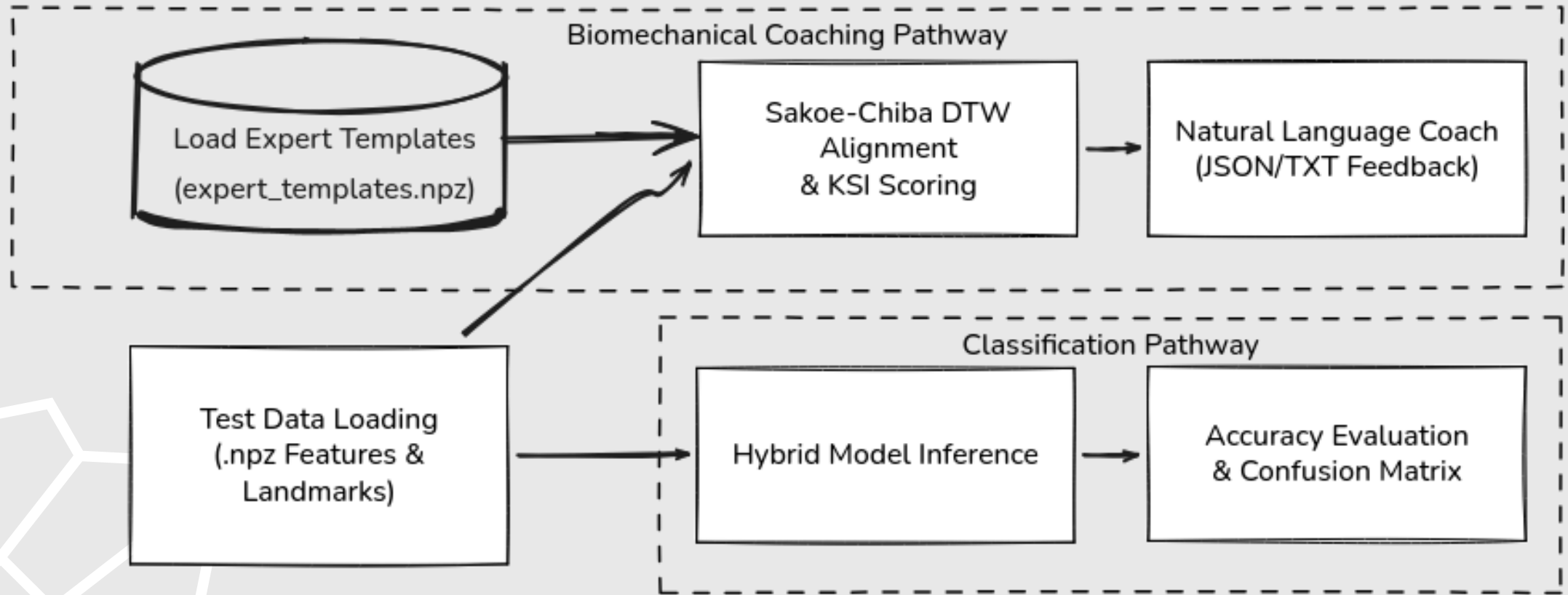
One-line takeaway: KSI not only scores motion quality, it pinpoints *where* the fault happens and turns it into clear, personalized coaching language.

Testing and Validation

- **Data Export & Pre-processing:** Expert videos are processed to extract raw landmarks and features.
- **DTW Alignment:** Uses **Dynamic Time Warping** to align multiple expert clips in time, ensuring disparate speeds of execution are synchronized.
- **Quality-Weighted Sequence:** A master template is generated by averaging aligned expert performances, creating a robust reference for scoring user errors.



Results and output



Relevant Dataset

Access: Videos from BWF Indonesia Open

The "badminton_storke_video" dataset offers a collection of badminton stroke videos categorized into six distinct actions: **forehand drive**, **forehand lift**, **forehand net shot**, **forehand clear**, **backhand net shot**, and **backhand clear**.

Forehand Clear



Forehand Drive



Forehand Lift



Forehand Net Shot



Backhand Drive



Backhand Net Shot



Demo Video

Badminton Analytics Pro

AI-Powered Shot Classification & Biomechanics Analysis

Analyze Video



No file chosen

Click or Drag Video Here

Supports MP4, MOV, AVI (Max 50MB)



25°C
Mostly clear



Search



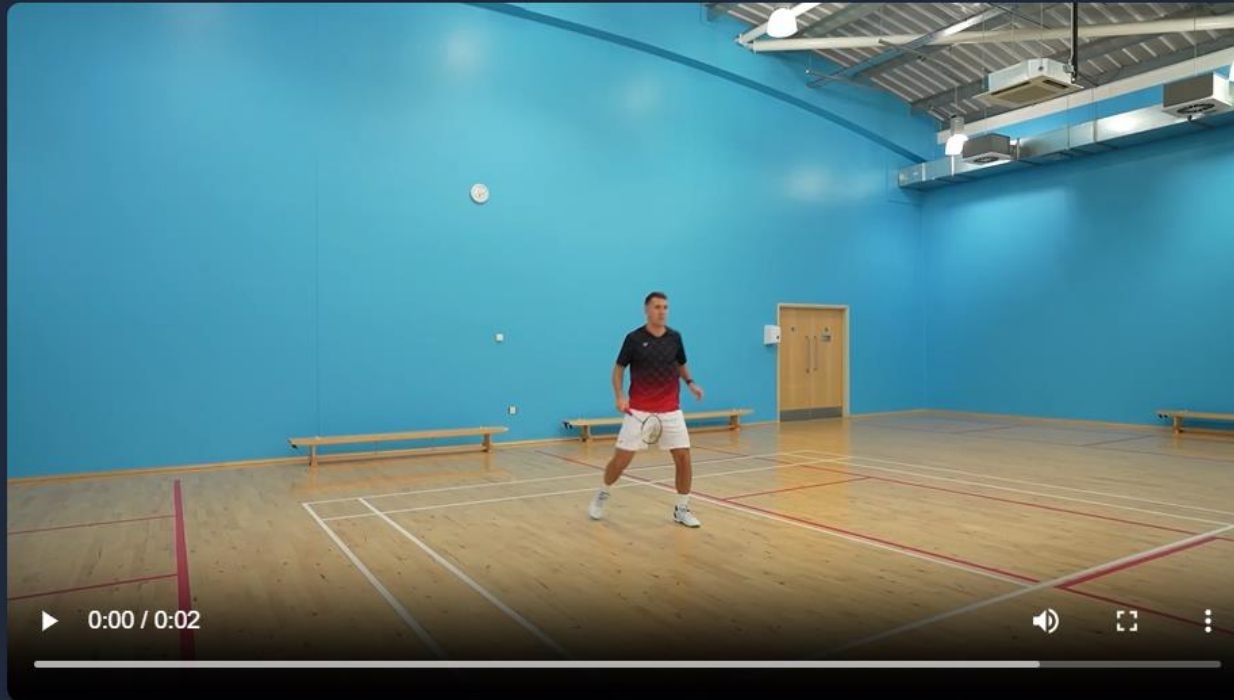
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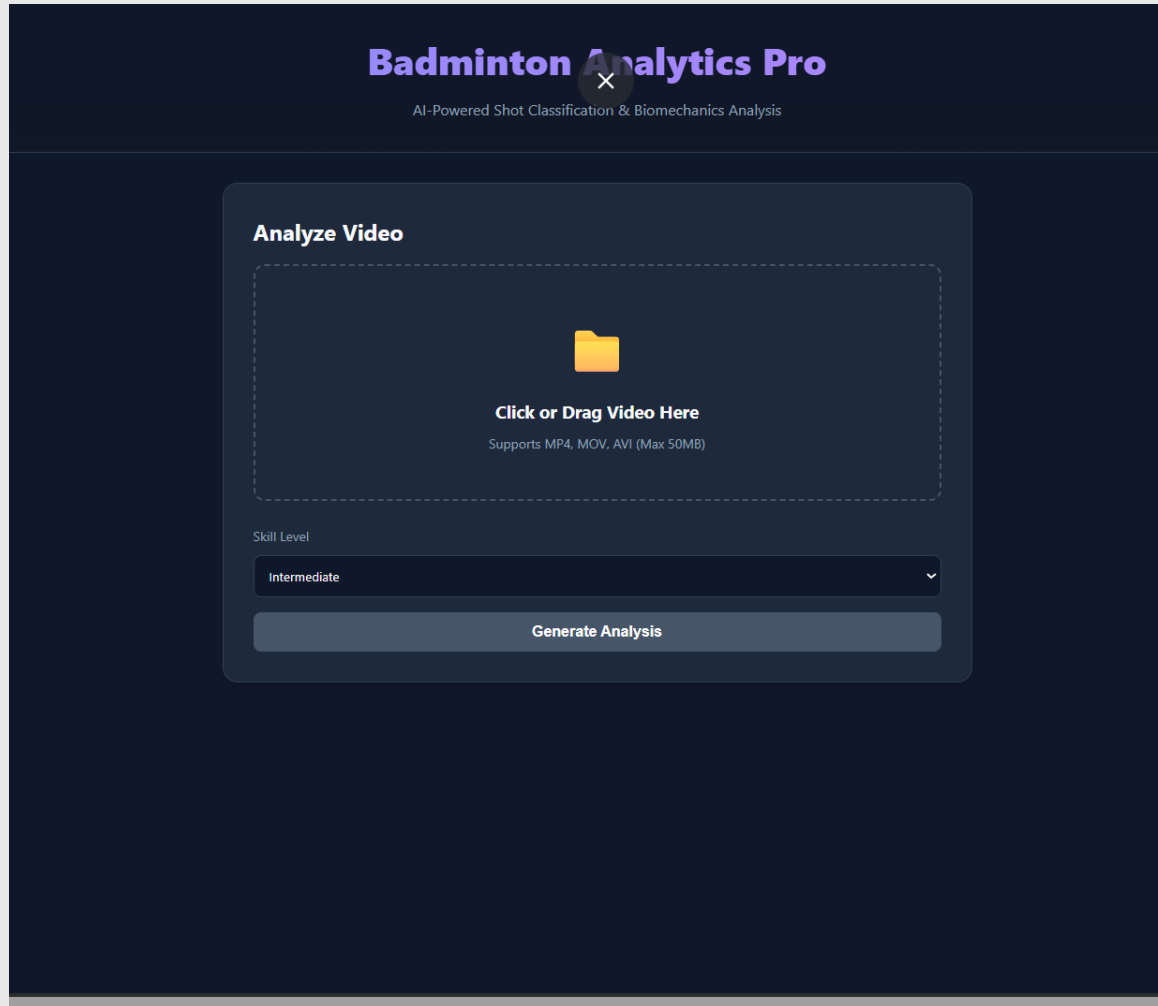
Correction Demo

Correct Execution (Slow Motion)

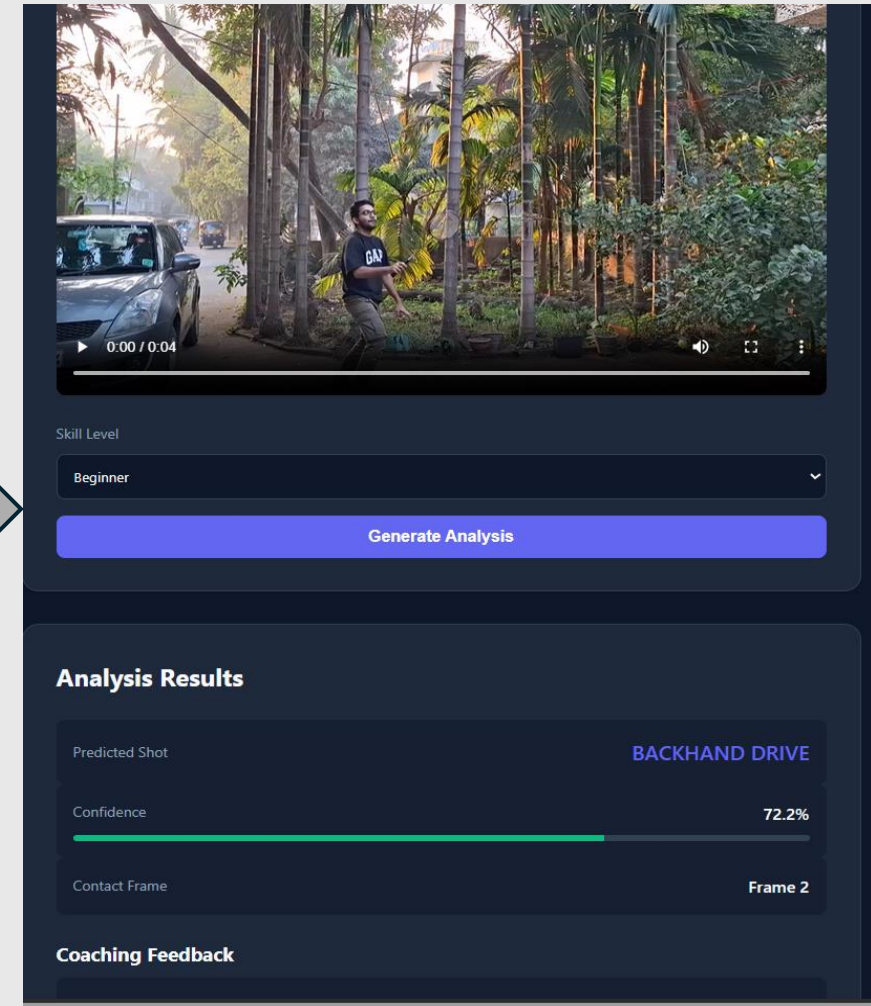
Watch the expert perform the **FOREHAND CLEAR**.



Test Cases



System interface for video file upload.



Real-time inference engine processing and keypoint extraction.

Test Cases

Analysis Results

Predicted Shot

BACKHAND DRIVE

Confidence

72.2%

Contact Frame

Frame 2

Coaching Feedback

★ Foundation Building

Let's rebuild your **Backhand Drive** from the ground up. 🙌

Your score of **0.34** (★ Foundation Building) indicates fundamental technique issues, but that's actually good news - it means small changes will create big improvements! Focus on the basics below, and you'll see rapid progress.

Priority Fixes

Issue: Your Spine Lateral Lean shows a 178.2° deviation from optimal technique.

Fix: Focus on your Spine Lateral Lean during practice. Use video analysis to compare with expert footage.

Drill: Practice the movement slowly, focusing on Spine Lateral Lean position. Use a mirror or video for feedback.

Issue: **Trunk Posture** significantly shows too much forward lean (170.2° deviation)

Fix: Keep your chest up and back relatively straight. Don't bend at the waist to reach - move your feet instead.

Drill: **Posture Control Drill:** Wall alignment: stand with back to wall, maintain contact through swing motion. Builds awareness. 2 sets of 10. *Duration: 5 minutes before each

Correct Execution (Slow Motion)

Watch the expert perform the **BACKHAND DRIVE**.

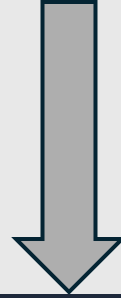
Technique & Exercises

FAST DRIVE

Expert reference demonstration for the Backhand Drive.

Shot classification and confidence metrics.

Test Cases



Error Visualization



Detected Errors:

- [00:02.86] Frame 86: Fix spine lateral lean
- [00:02.86] Frame 86: Fix spine forward lean
- [00:02.86] Frame 86: Fix shoulder rotation angle
- [00:02.86] Frame 86: Fix hip rotation angle
- [00:02.86] Frame 86: Fix right knee angle

Biomechanical error detection and personalized coaching feedback.

Result Analysis

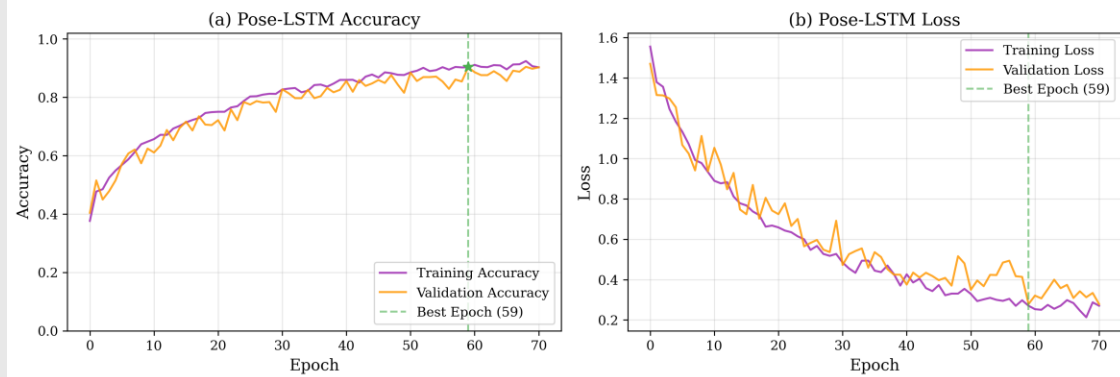
Shot Class	Precision	Recall	F1-Score	Support
Backhand Drive	0.96	1.00	0.98	46
Backhand Net Shot	0.98	1.00	0.99	58
Forehand Clear	1.00	1.00	1.00	48
Forehand Drive	1.00	0.95	0.98	42
Forehand Lift	1.00	0.98	0.99	42
Forehand Net Shot	0.98	0.98	0.98	42
Macro Average	0.99	0.98	0.98	278

Method	Sport	Accuracy	Feedback	Input
Siddiqui et al. [11]	Cricket	99.7%	None	Skeleton
Harini et al. [21]	Squat Exercise	91.0%	Rule-based	Skeleton
Chen et al. [26]	General Sports	–	DTW-based	Skeleton
Smashifix (Proposed)	Badminton	98.6%	KSI + DTW	Skeleton + Visual

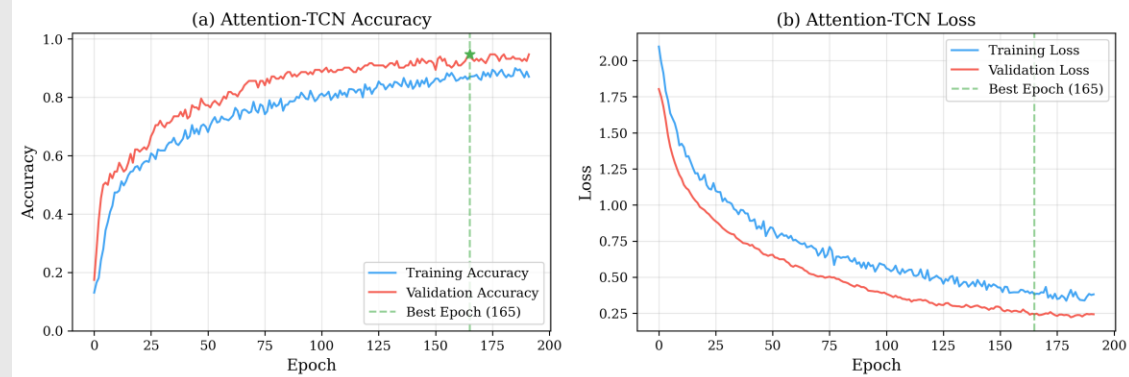
Model Architecture	Accuracy	Precision	Recall	F1-Score
Pose-LSTM (Real-time)	0.81	0.79	0.78	0.78
Attention-TCN (Proposed)	0.9856	0.99	0.98	0.98
RSN (Visual only)	0.74	0.72	0.71	0.71

Result Analysis

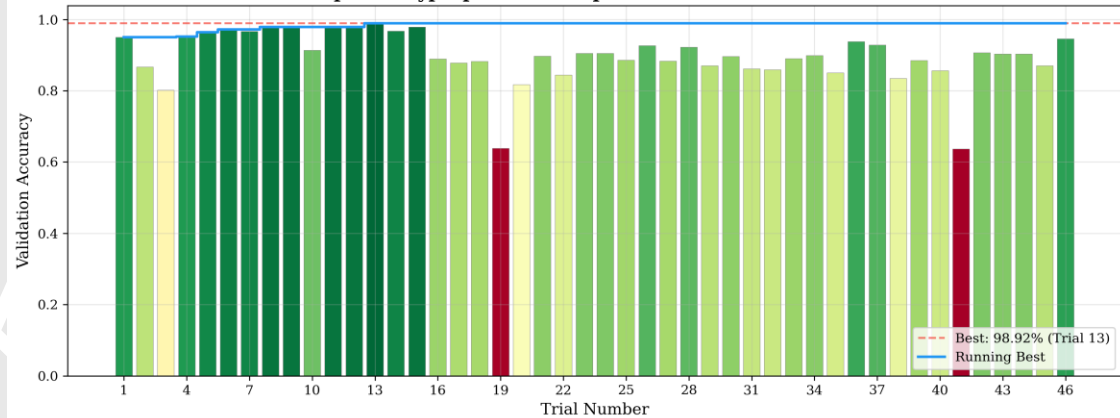
Pose-LSTM Training and Validation Curves



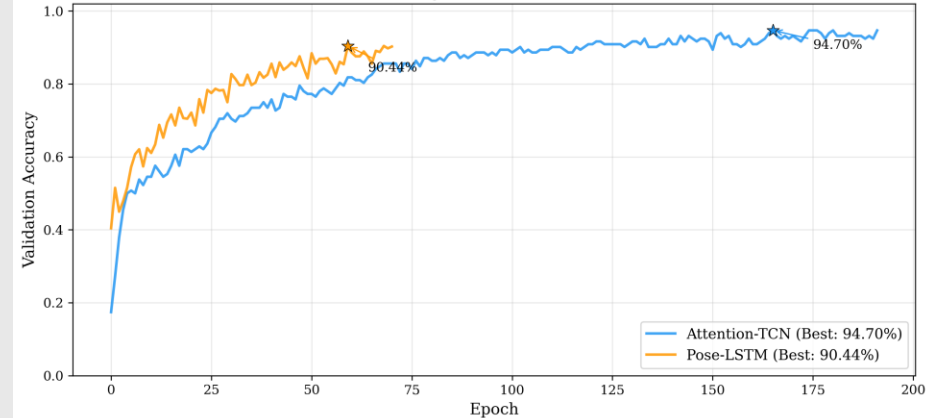
Attention-TCN Training and Validation Curves



Optuna Hyperparameter Optimization: Attention-TCN



Validation Accuracy: Attention-TCN vs Pose-LSTM



Conclusion & Future Scope

This project successfully developed Smashifix, an automated badminton coaching system that bridges the gap between amateur practice and professional biomechanical analysis. By utilizing a dual-pipeline architecture, the system delivers both ultra-low latency real-time feedback on the court and high-fidelity forensic analysis, achieving up to 98.92% validation accuracy. The integration of a mathematically rigorous scoring system with a Natural Language Coach ensures that complex movement data is seamlessly translated into actionable, personalized training drills. Ultimately, Smashifix provides an optimal balance of classification accuracy and feedback granularity, setting a strong foundation for future advancements like edge-device deployment and multi-player tracking.



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